RPMGLOBAL

AMT - Scheduling Methods

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. This guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

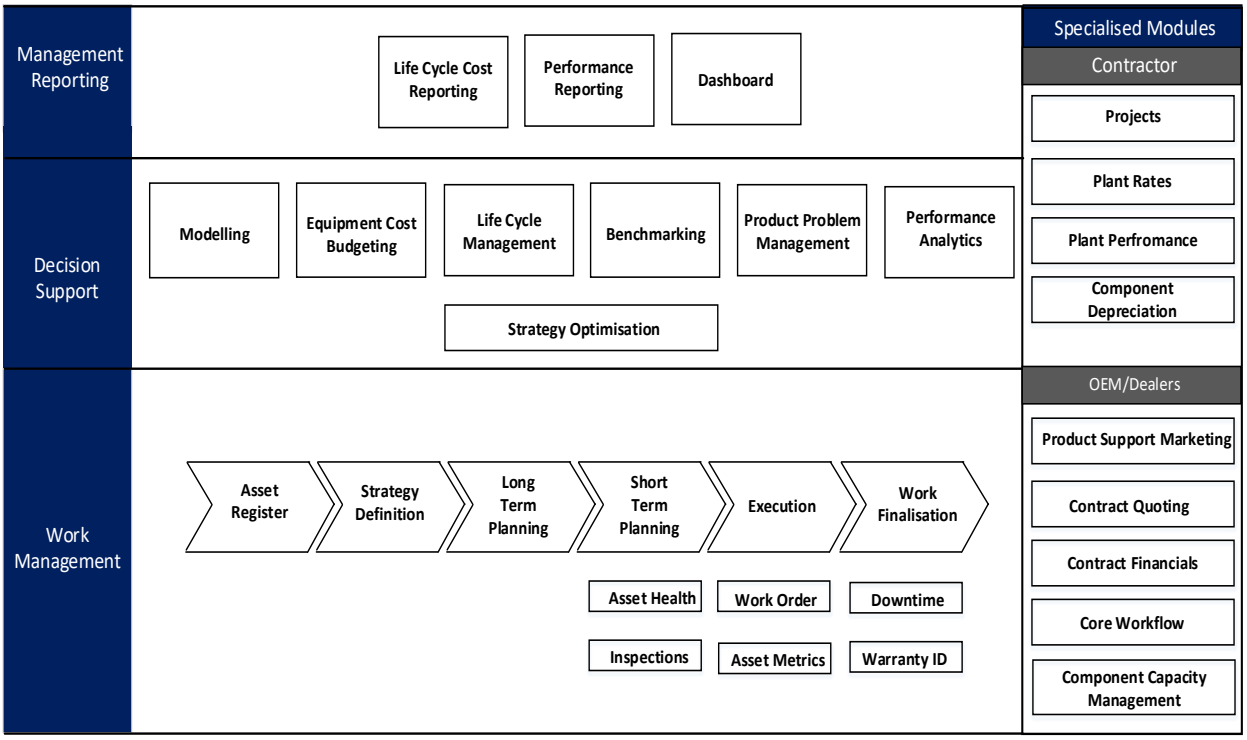
Check your understanding on the workshop and ask any related questions.

Course Outline

Prerequisites	
Target Audience	
Nominal Duration	
Course Objectives	
Learning Outcomes	At the end of the course, participants will have learned to: •
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below.



INTRODUCTION

Depending on the nature of scheduling required by a site for particular strategy tasks (for example, PM services), the tasks could be set up using one of the following logics available in AMT:

- Standard
- Dependency
- Suppression (parent or sibling)
- Inspection

As well as basing the strategy tasks on one of the above logics, you can also choose whether the next occurrence of the task is based on either:

- Last Performed UOM / Date
- Last Scheduled UOM / Date

The above concepts will be explained in the remainder of this handbook.

Training Workshops

1. Business Principles

Before considering whether suppression or dependency logic is required, you must first understand the different options available for determining how the next occurrence of a strategy task can be forecast.

When initially setting up strategy tasks in a projection, you must choose whether to base the next occurrence of that task on either:

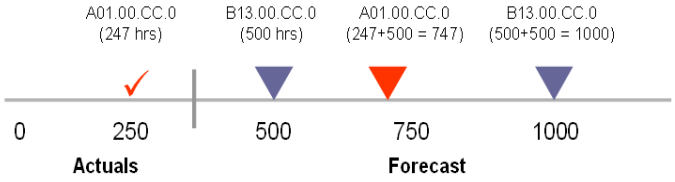
- Last Performed UOM/Date
- Last Scheduled UOM/Date

The following two sections explain each of these options.

LAST PERFORMED

Last Performed means that the forecasted next occurrence of the strategy task is based on the last time the strategy task was actually executed. Below is a simplified example.

Example:	First	Frequency	Scheduling Logic
A01.00.CC.0	250	500	Last Performed
B13.00.CC.0	500	500	Last Performed



In the above example, the projection is set up with the A1.00.CC.0 task forecast to be first performed at 250 hours and the next task at 500 hours after. Therefore, the second occurrence of the A1.00.CC.0 task is forecast to occur at 750 hours.

However, A1.00.CC.0 is actually performed on the equipment at 247 hours (3 hours earlier than planned). Therefore, because the strategy task is based on the Last Performed UOM, the next occurrence is forecast by AMT to occur at 747 hours (that is, 247 + 500). In other words, AMT takes into consideration when the task was actually performed when forecasting the next occurrence.

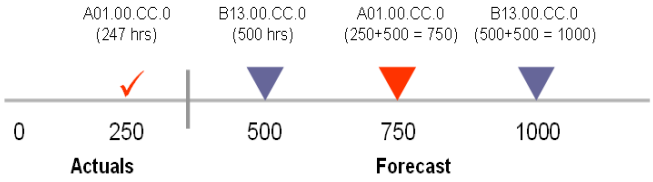


*AMT still requires a work order to be linked to the strategy task executed, otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of whether Last Performed or Last Scheduled logic is used).*

LAST SCHEDULED

Last Scheduled means that the forecasted next occurrence of the strategy task is based on when the strategy task was scheduled to occur. It ignores the usage or date when the job was performed. See the example below.

Example:	First	Frequency	Scheduling Logic
A01.00.CC.0	250	500	Last Scheduled
B13.00.CC.0	500	500	Last Scheduled



In the above example, the projection is set up with the A1.00.CC.0 task forecast to be first performed at 250 hours and the next task 500 hours after. Therefore, the second occurrence of the A1.00.CC.0 task is forecast to occur at 750 hours.

The A1.00.CC.0 task is then performed on the equipment at 247 hours (3 hours earlier than planned). However, because the strategy task is based on the Last Scheduled UOM, the next occurrence is forecast by AMT to occur at 750 hours (that is, 250 + 500). In other words, AMT ignores when the task actually occurred when forecasting the next occurrence.



*AMT still requires a work order to be linked to the strategy task executed, otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of whether Last Performed or Last Scheduled logic is used).*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

2. Scheduling – Standard

Standard scheduling means that the strategy task relies on the First, Frequency and Not After parameters you enter when editing a strategy task.

Consequently, forecasting future occurrences of the strategy task is not influenced by the execution of other strategy tasks (unlike suppression and dependency logics). AMT treats the strategy task in isolation.

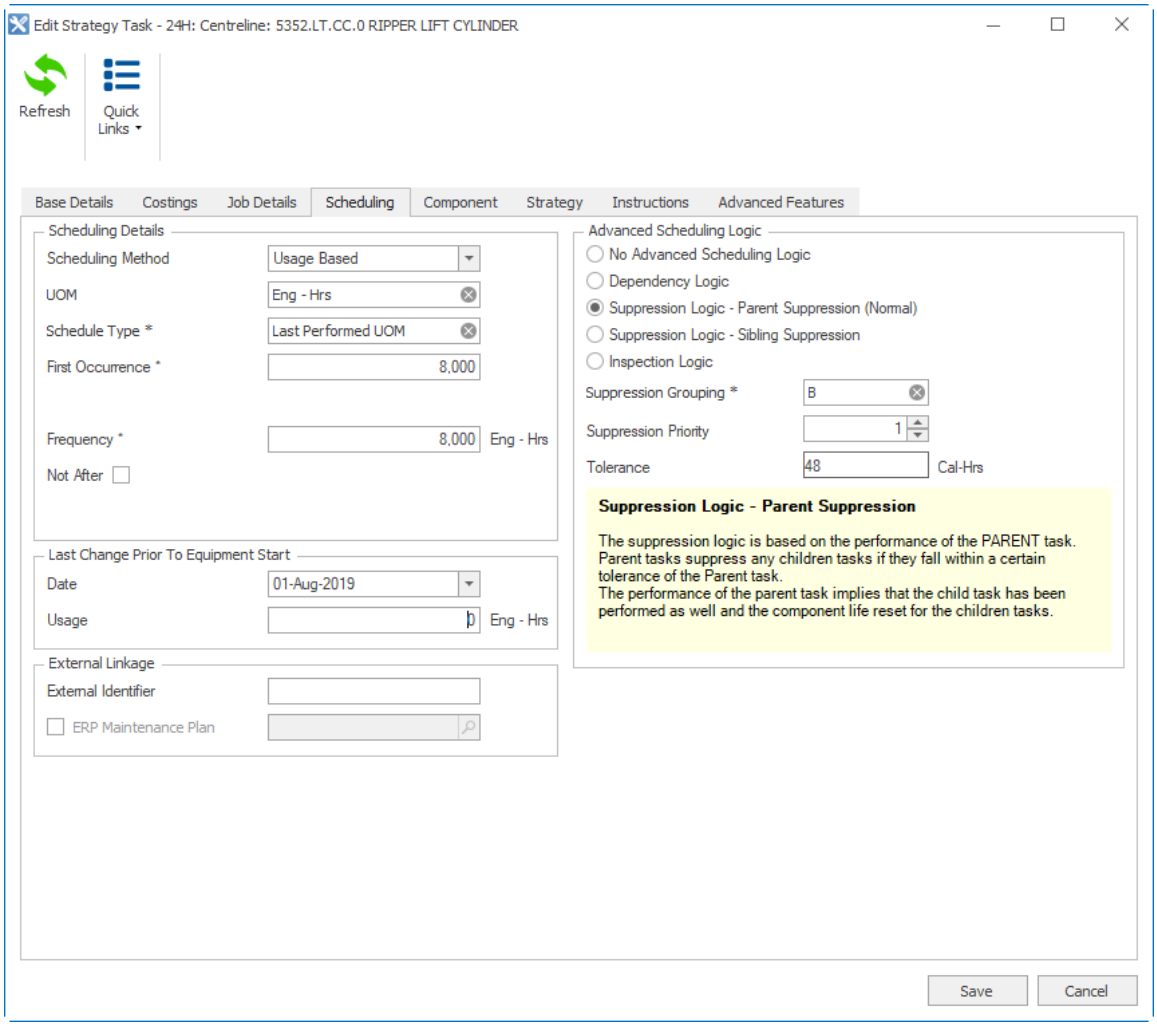
Typically, most strategy tasks are set up with the standard form of scheduling.

3. Scheduling Parameters – Parent Suppression

Procedure – Set Up Parent Suppression

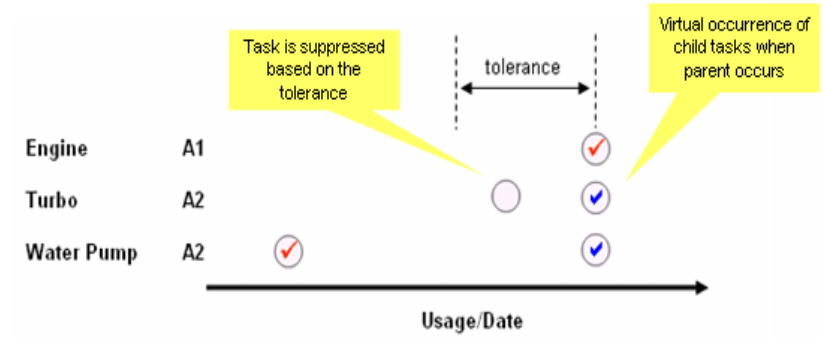
To set up suppression for a task:

1. Locate and edit the task in the equipment's projection.
2. Click the **Scheduling** tab.
3. Select **Suppression Logic – Parent Suppression (Normal)**.
4. Select a **Suppression Grouping**.
5. In the **Scheduling Details** panel, select a **Suppression Priority**.
6. Set the **First Occurrence** and **Frequency** parameters.
7. Click **Save**.

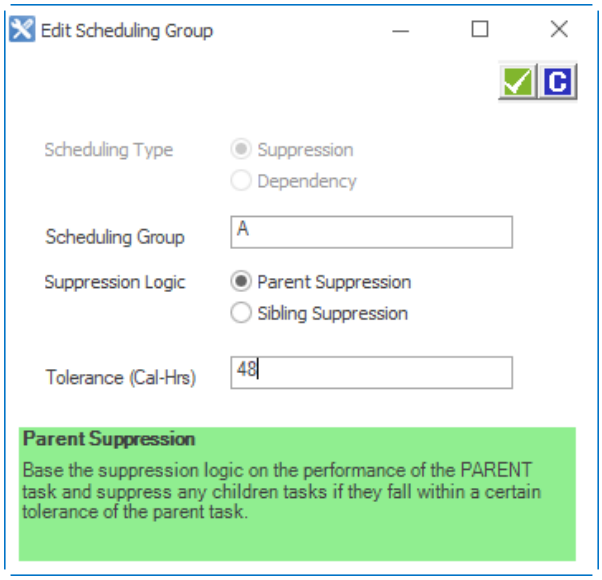


Tasks suppress another task when it falls within a certain tolerance; the default tolerance is 48 calendar hours. If the child task falls within the tolerance of the parent task, the child task is suppressed and the parent task is performed.

The parent task within the suppression group should have the LOWEST number in the group.

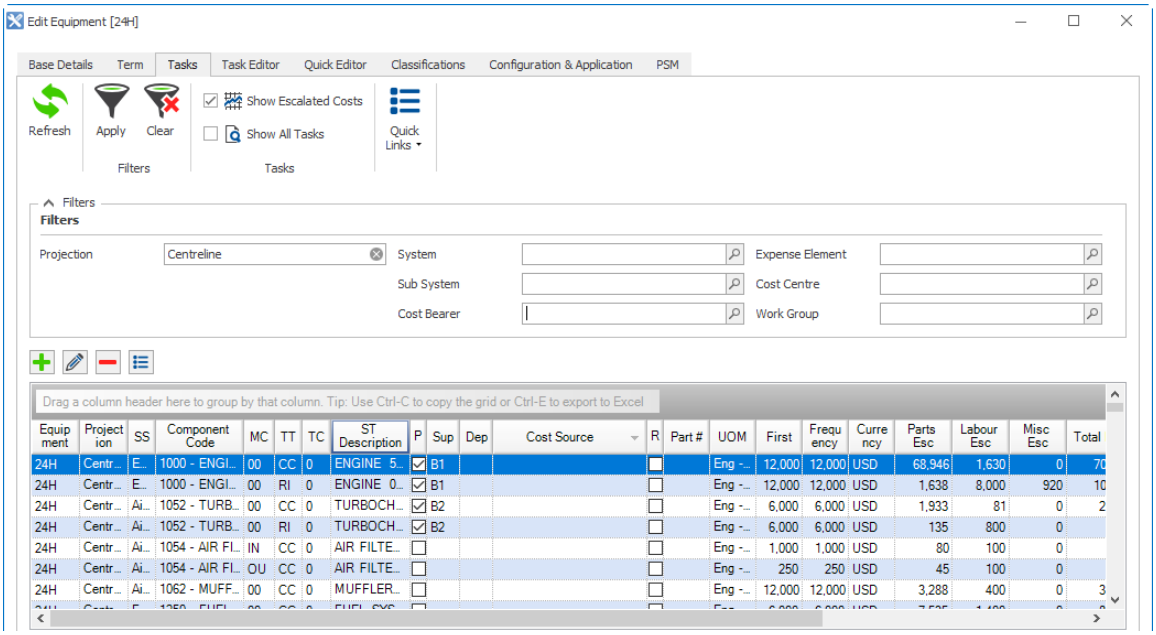


Tolerances for each scheduling logic type (that is, dependency, suppression and so on) are maintained in the system table **Scheduling Group**.



It is possible to have more than one task with the same “grouping + sequence” combination. For example, you may want to have an engine as a strategy task as well as its sub-components, turbo and radiator. Using suppression, the set up would be:

1. Engine (A1)
2. Turbo (A2)
3. Radiator (A2)



Consequently, when the engine task occurs, AMT resets the life of the turbo and radiator.



*AMT still requires a work order to be linked to the strategy task that was executed, otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of its scheduling logic).*

See the Attachments section of this document for another example of how PM services can be set up using parent suppression.

Learning Check



Check your understanding of the workshop and ask any related questions.

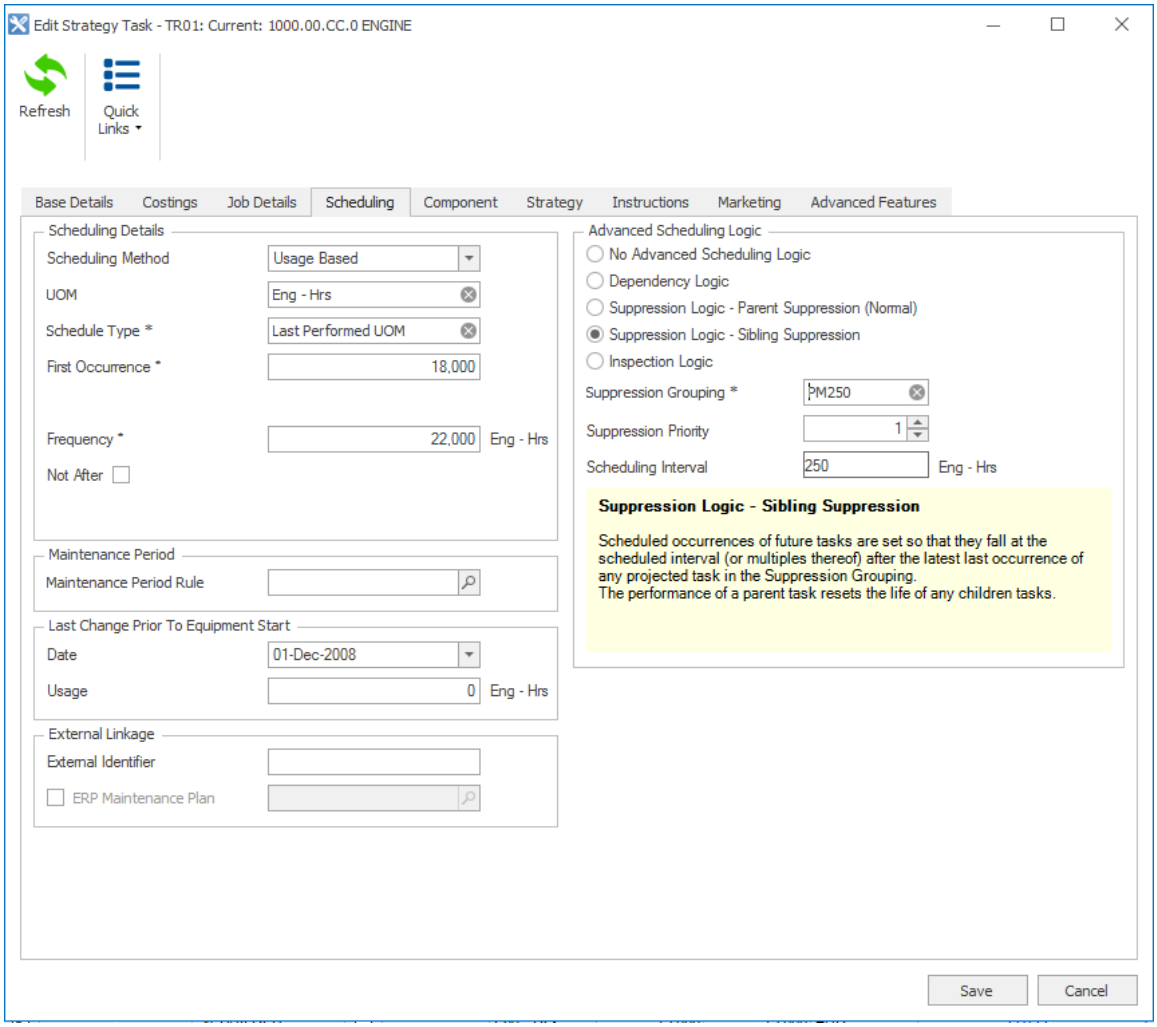
Notes

4. Scheduling Parameters – Child Suppression

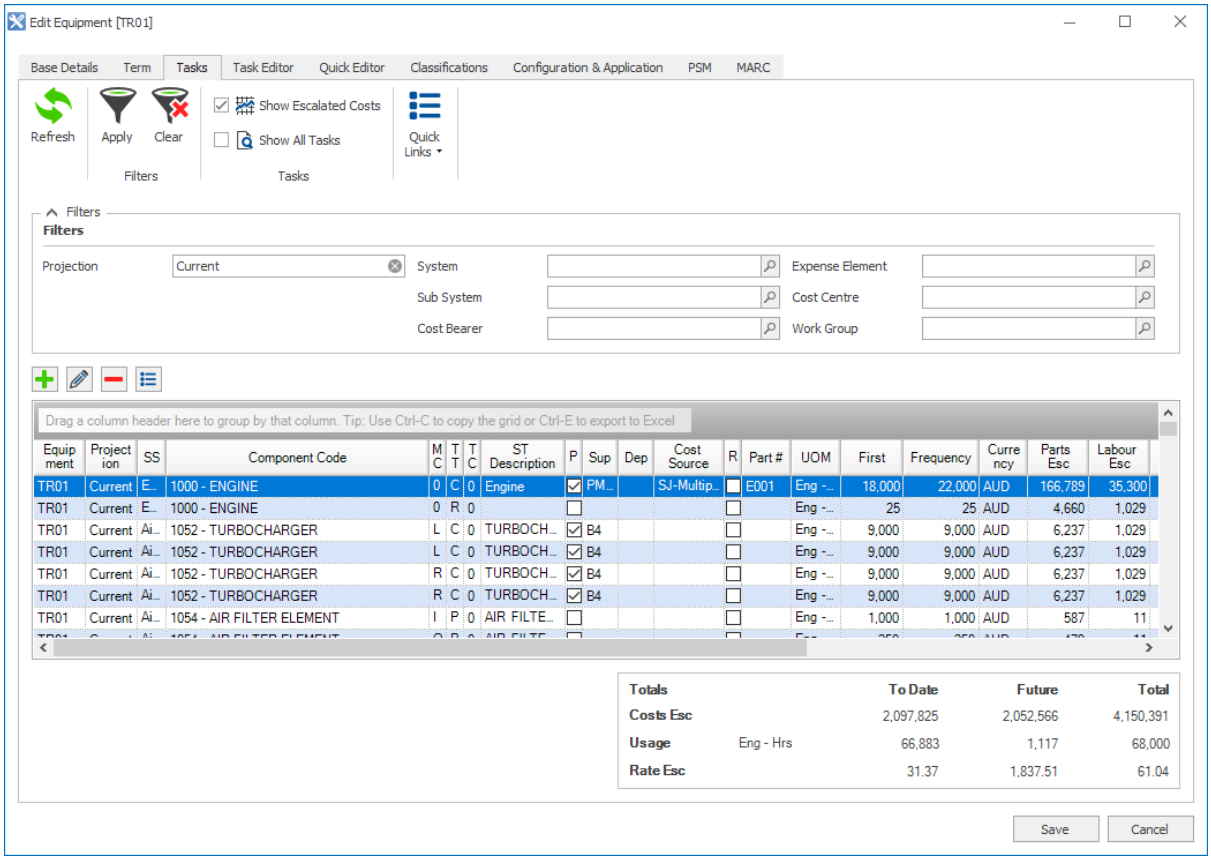
Procedure – Set Up Sibling Suppression

To set up sibling suppression for a task:

1. Locate and edit the task in the equipment's projection.
2. Click the **Scheduling** tab.
3. Select **Suppression Log – Sibling Suppression**.
4. Select a **Grouping**.
5. Select a **Sequence number**.
6. Set **First Occurrence** and **Frequency** parameters.
7. Click **Save**.



The parent task within the suppression group should have the LOWEST number in the group.



*AMT still requires a work order to be linked to the strategy task that was executed, otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of its scheduling logic).*

Learning Check



Check your understanding of the workshop and ask any related questions.

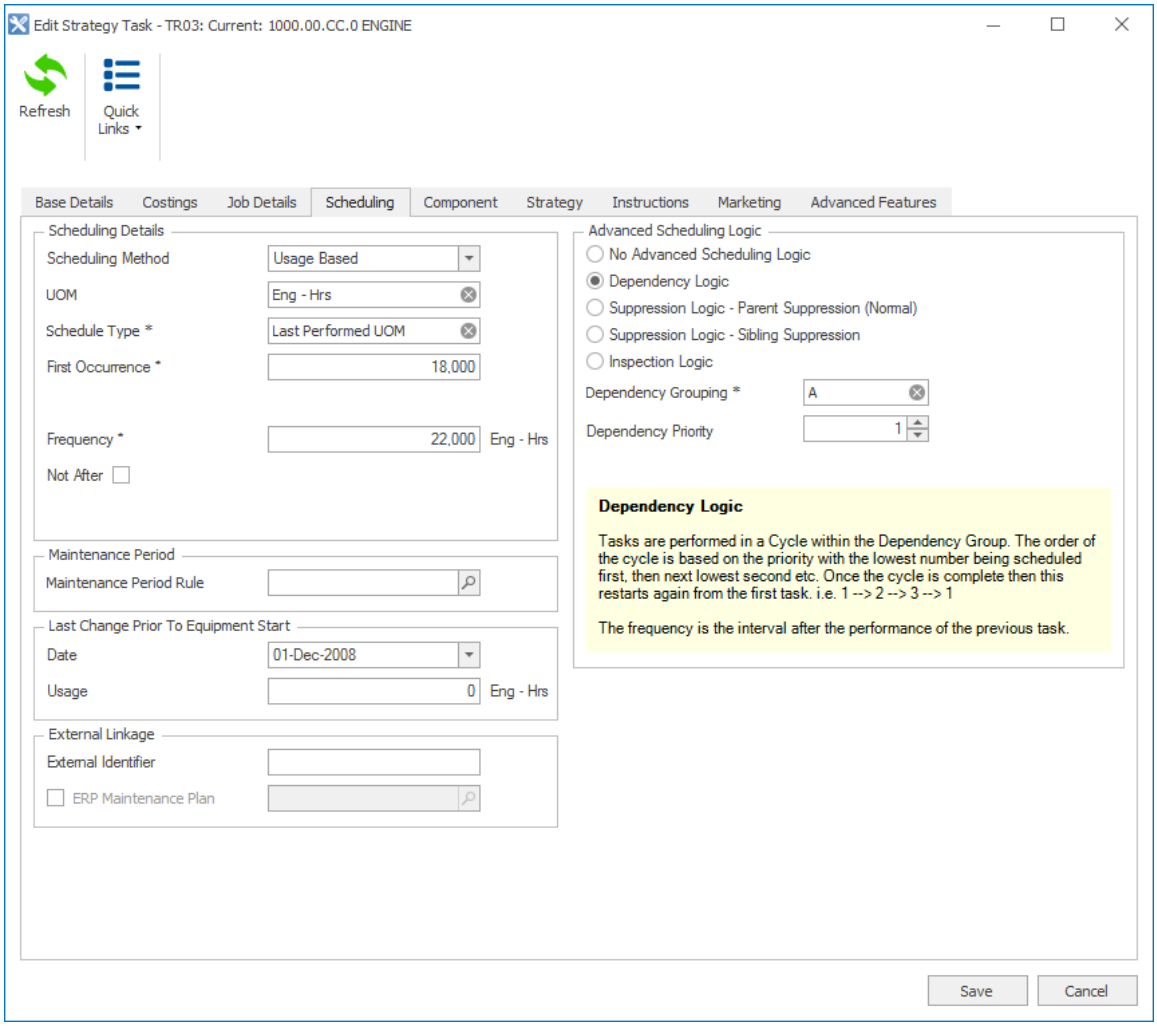
Notes

5. Scheduling – Dependency

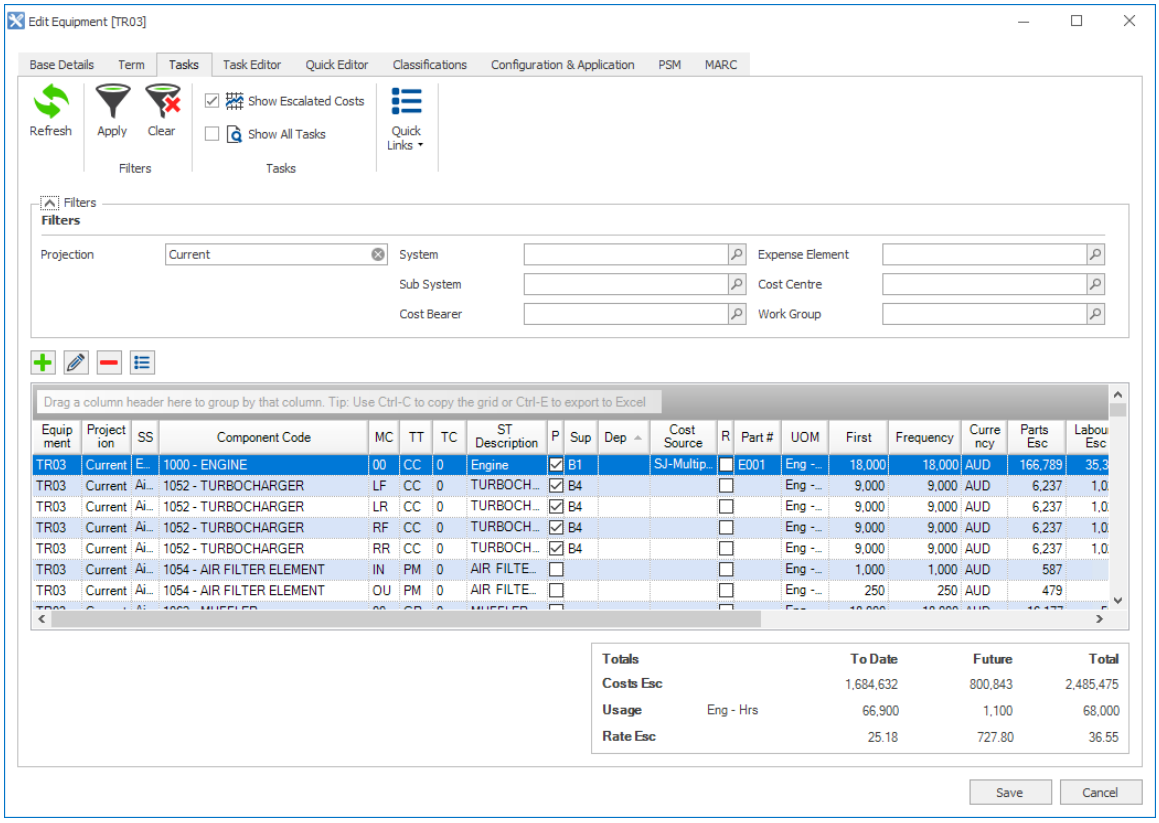
Procedure

To set up dependency for a task:

1. Locate and edit the desired task in the equipment's projection.
2. Click the **Scheduling** tab.
3. Select **Dependency Logic**.
4. Select a **Grouping**.
5. Select a **Sequence number**.
6. Set **First** and **Frequency** parameters.
7. Click **Save**.



- The Group (for example, A) determines which tasks are related to each other.
- The Number (for example, 1,2,3) determines the order, whether the lowest number occurs first.
- The Frequency (for example, 500) is the interval after the previous task.



See the Attachment section of this document for another example of how PM services can be set up using dependency.

Learning Check



Check your understanding of the workshop and ask any related questions.

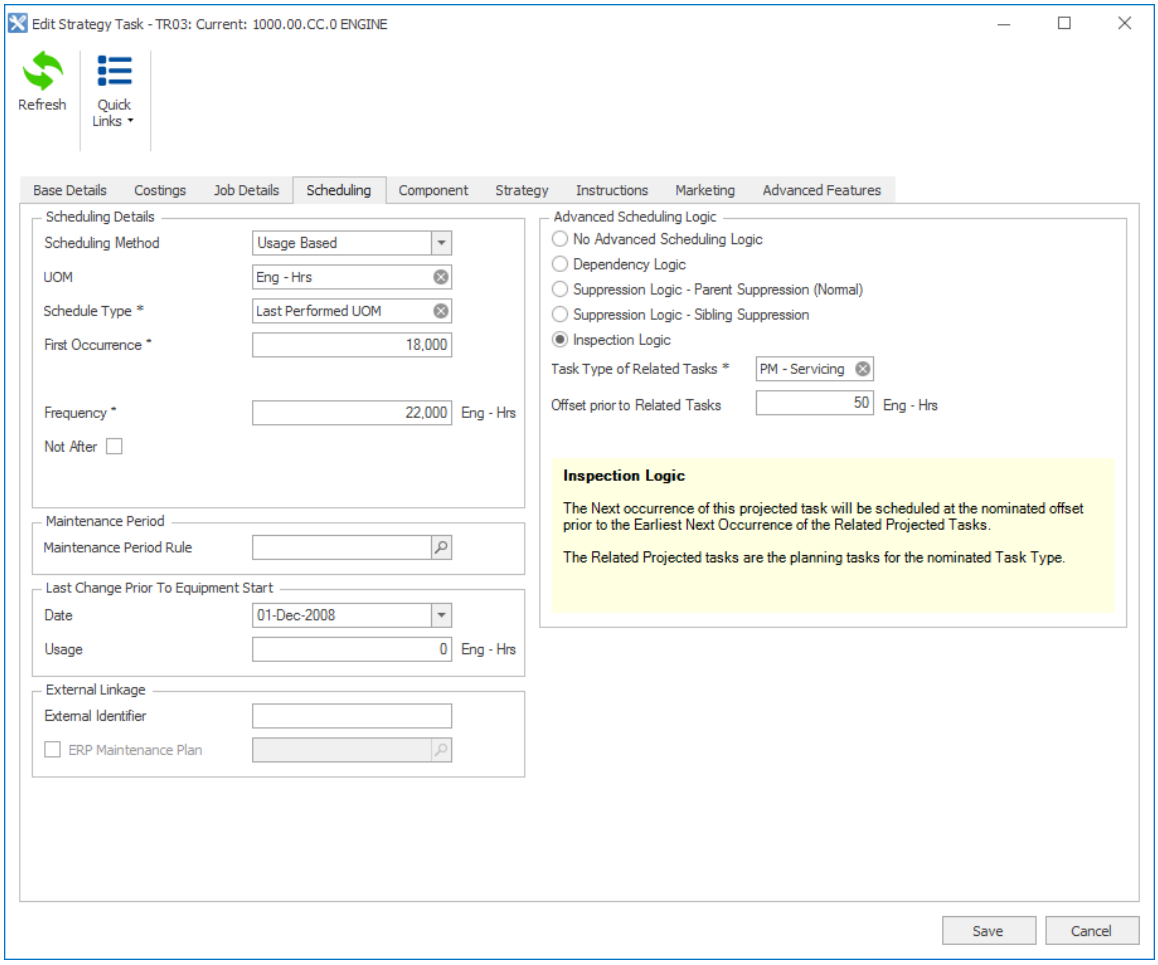
Notes

6. Scheduling – Inspection

Procedure – Set Up Inspection Logic

To set up inspection logic for a task:

1. Locate and edit the task in the equipment's projection.
2. Click the **Scheduling** tab.
3. Select **Inspection Logic**.
4. Select an **Offset prior to Related Tasks**.
5. Click **Save**.



- **Task Type of Related Tasks** identifies which projected tasks this task should precede.
- **Offset prior to Related Tasks** determines by how many hours or days this task will precede the nominated projected tasks (as identified by **Task Type of Related Tasks**).

*Manually rescheduling an Inspection Logic task in Strategy Confirmation or in the **Next Due** tab in the projected task will not adjust the next occurrence of the projected task to which the inspection task is related.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

7. Summary

Strategy tasks can have special scheduling logic applied:

- Last Performed or Last Scheduled
- Standard
- Suppression – Parent
- Suppression – Sibling
- Dependency
- Inspection

Based on the logic chosen, AMT automatically sends the next strategy task in the scheduling sequence to the Planning module. You can see which scheduling logic has been set up for an equipment by opening the equipment's current projection.

If you do not have access to current projections, contact your AMT Administrator.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

A. Attachments



Example – Parent Suppression for PMs

Functionality Overview

The images below show how PM services should be set up in AMT using suppression.

TASK SUMMARY

Drag a column header here to group by that column. Tip: Use Ctrl-C to copy the grid or Ctrl-E to export to Excel

Equipment	Projection	SS	Component Code	MC	TT	TC	ST Description	P	Sup	Dep	Cost Source	R	Part #	UOM	First	Frequency	Currency	Parts Esc	Labour Esc	Misc
TR02	Curre...	Engine	1000 - ENGI...	00	CC	0	Engine	☑	B1		SJ-Multi...	☐	E001	Eng -...	18,000	22,000	USD	166,789	35,300	
TR02	Curre...	Air Inducti...	1052 - TURB...	LF	CC	0	TURBOCH...	☑	B4		SJ-BUIL...	☐		Eng -...	9,000	9,000	USD	1,713	1,000	
TR02	Curre...	Air Inducti...	1052 - TURB...	LR	CC	0	TURBOCH...	☑	B4			☐		Eng -...	9,000	9,000	USD	6,237	2,585	
TR02	Curre...	Air Inducti...	1052 - TURB...	RF	CC	0	TURBOCH...	☑	B4			☐		Eng -...	9,000	9,000	USD	6,237	2,585	
TR02	Curre...	Air Inducti...	1052 - TURB...	RR	CC	0	TURBOCH...	☑	B4			☐		Eng -...	9,000	9,000	USD	6,237	2,585	
TR02	Curre...	Cooling S...	1355 - THER...	00	CC	0	Thermostats	☑	B4			☐		Eng -...	6,000	6,000	USD	274	500	
TR02	Curre...	Cooling S...	1359 - FAN...	00	CC	0	FAN DRIVE	☑	B4			☐		Eng -...	9,000	9,000	USD	1,469	1,034	
TR02	Curre...	Cooling S...	1361 - WAT...	00	CC	0	WATER P...	☑	B4			☐		Eng -...	9,000	9,000	USD	1,230	750	
TR02	Curre...	Cooling S...	1363 - AFTE...	00	CC	0		☑	B4			☐		Eng -...	9,000	9,000	USD	1,148	750	
TR02	Curre...	Starting S...	1451 - AIR S...	00	CC	0	AIR START...	☑	B4			☐		Eng -...	9,000	9,000	USD	3,967	646	
TR02	Curre...	Engine	1803 - AIR C...	00	CC	0	AIR COMP...	☑	B4			☐		Eng -...	9,000	9,000	USD	2,866	646	
TR02	Curre...	PM	7527 - 2000...	00	PM	0	PM8	☑	PM2501		SJ-BUIL...	☐		Eng -...	2,000	2,000	USD	1,692	1,600	
TR02	Curre...	PM	7526 - 1000...	00	PM	0	PM4	☑	PM2502		SJ-BUIL...	☐		Eng -...	1,000	1,000	USD	1,692	1,200	
TR02	Curre...	PM	7525 - 500 S...	00	PM	0	PM2	☑	PM2503		SJ-BUIL...	☐		Eng -...	500	500	USD	932	800	
TR02	Curre...	PM	7524 - 250 S...	00	PM	0	PM1	☑	PM2504		SJ-BUIL...	☐		Eng -...	0	250	USD	294	400	

The hierarchy is that A1 is the parent of A2 is the parent of A3 is the parent of A4, and so on. So, when A1 occurs, the life for all of its children is reset. 2000 and 3000 hours services are at the same level = A2. They both suppress the 1000 service.

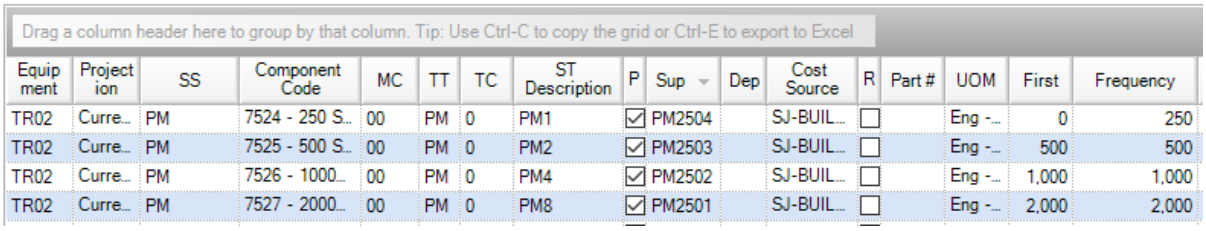
RESULTANT FORECAST (COMPONENT SCHEDULE REPORT)

Component	Part No	R	Rotable Serial No	Mod Cod	Task Type	Occ Type	Strategy Locked	O'due	Due Date	Usage	Life Used	UOM	Proj Interval	Proj Cost	Proj Lab Hrs	Proj Duration
Machine: RF_01																
7524 250 SVC HOUR MAINTEN				00	PM				31-Dec-06	250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				30-Jan-07	500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-Mar-07	750	26	Eng Hr	250	250	0	0
7526 1000 SVC HOUR MAINTE				00	PM				01-Apr-07	1,000	26	Eng Hr	1,000	1,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-May-07	1,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				01-Jun-07	1,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-Jul-07	1,750	26	Eng Hr	250	250	0	0
7527 2000 SVC HOUR MAINTE				00	PM				01-Aug-07	2,000	26	Eng Hr	2,000	2,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Aug-07	2,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				01-Oct-07	2,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Oct-07	2,750	26	Eng Hr	250	250	0	0
7545 3000 SVC HOUR MAINTE				00	PM				01-Dec-07	3,000	26	Eng Hr	3,000	3,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Dec-07	3,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				31-Jan-08	3,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-Mar-08	3,750	26	Eng Hr	250	250	0	0
7527 2000 SVC HOUR MAINTE				00	PM				01-Apr-08	4,000	26	Eng Hr	2,000	2,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-May-08	4,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				31-May-08	4,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-Jul-08	4,750	26	Eng Hr	250	250	0	0
7526 1000 SVC HOUR MAINTE				00	PM				31-Jul-08	5,000	26	Eng Hr	1,000	1,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Aug-08	5,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				30-Sep-08	5,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Oct-08	5,750	26	Eng Hr	250	250	0	0
7543 6000 SVC HOUR MAINTE				00	PM				30-Nov-08	6,000	26	Eng Hr	6,000	6,000	0	0

Example – Dependency for PMs

Functionality Overview

The image below illustrates how PM services using dependency logic should be set up.



Drag a column header here to group by that column. Tip: Use Ctrl-C to copy the grid or Ctrl-E to export to Excel

Equipment	Projection	SS	Component Code	MC	TT	TC	ST Description	P	Sup	Dep	Cost Source	R	Part #	UOM	First	Frequency
TR02	Curre...	PM	7524 - 250 S...	00	PM	0	PM1	☑	PM2504		SJ-BUIL...	☐		Eng -...	0	250
TR02	Curre...	PM	7525 - 500 S...	00	PM	0	PM2	☑	PM2503		SJ-BUIL...	☐		Eng -...	500	500
TR02	Curre...	PM	7526 - 1000...	00	PM	0	PM4	☑	PM2502		SJ-BUIL...	☐		Eng -...	1,000	1,000
TR02	Curre...	PM	7527 - 2000...	00	PM	0	PM8	☑	PM2501		SJ-BUIL...	☐		Eng -...	2,000	2,000

The PM tasks follow each other in a sequence - A1, then A2 then A3 then A4 then back to A1. Whenever the last task was performed, the next is scheduled based on the last performed.

Hypothetically, if PM services were performed in the past, we can identify that PM 2 was performed at 39,652 hours (which can be confirmed by looking at the work order history using **Workorder Settlement Manager**).

Consequently, AMT will calculate the other PM services in the dependency group to occur at:

- PM 3 at 40,152 hours
- Followed by a PM 4 at 40,652 hours
- Followed by a PM 1 at 41,152 hours
- Followed by a PM 2 at 41,652 hours

RESULTANT FORECAST (COMPONENT SCHEDULE REPORT)

Component	Part No	R	Rotable Serial No	Mod Cod	Task Type	Occ Type	Strategy Locked	O'due	Due Date	Usage	Life Used	UOM	Proj Interval	Proj Cost	Proj Lab Hrs	Proj Duration
Machine: TR002																
7503 PM 3				00	PM	SRV			15-Nov-06	40,152	1,049	Eng Hr	500	2,056	72	12
7504 PM 4				00	PM				15-Dec-06	40,652	728	Eng Hr	500	742	6	1
7501 PM 1				00	PM				15-Jan-07	41,152	588	Eng Hr	500	1,047	24	6
7502 PM 2				00	PM				14-Feb-07	41,652	362	Eng Hr	500	1,732	60	10
7503 PM 3				00	PM				17-Mar-07	42,152	1,049	Eng Hr	500	2,056	72	12
7504 PM 4				00	PM				16-Apr-07	42,652	728	Eng Hr	500	742	6	1
7501 PM 1				00	PM				16-May-07	43,152	588	Eng Hr	500	1,047	24	6